

The Diversity of TCR α - β Chain Pairing

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April 2026

Abstract

T cell receptors (TCRs) mediate adaptive immunity via the specific binding of their $\alpha\beta$ -heterodimers to presented peptide antigens. While both the α - and β -chain is known to dictate antigen specificity, the majority of TCR repertoire studies and machine learning models rely exclusively on β -chain repertoires due to the throughput limitations of paired-chain sequencing. This study interrogates the inherent α - β -pairing promiscuity, which is neglected when β -chains are employed as proxies for TCRs, using paired-chain sequences curated from the Observed T cell receptor Space (OTS) database, the largest public paired-chain TCR database. Using a theoretical framework that considers both the distributional entropy and the functional divergence of α - β -pairings, the α -chain pairing diversity from the perspective of β -chains is quantified using four measures: the number of unique α -pairings, the perplexity of pairing, the Simpson index of pairing, and the expected Levenshtein distance between different α -chain partners. It is demonstrated that while most β -chains are only observed to pair with a single α -partner due to sparse sampling of TCR repertoires, pairing multiplicity is still observed to increase with sampling depth, both within and across repertoires. Furthermore, distinct α -chain partners of the same β -chain is shown to exhibit significant sequence differences between their complementarity-determining regions (CDRs), to a degree that would indicate differential epitope specificity. A proof-of-concept structural example demonstrates how the proposed diversity framework can be used to identify β -chains whose epitope specificity will likely vary by α -partner. These findings suggest that β -chains are imperfect proxies for TCRs, and that pairing ambiguity constitutes a non-negligible source of noise in single-chain repertoire models.

1 Introduction

The adaptive immune system must distinguish self from non-self. This is achieved through the high-specificity

binding of adaptive immune receptors to pathogenic antigens [1]. To cover the immense space of possible antigens, however, an enormous diversity of immune receptors is required [2].

T cells mediate adaptive immunity through the surface expression of T cell receptors (TCRs), which recognise intracellular or extracellularly-derived antigens [3–5]. Specifically, this recognition is achieved by the non-covalent binding of the TCR $\alpha\beta$ -heterodimer to linear antigenic peptides presented on peptide-major histocompatibility complexes (pMHC) [4, 5]. This makes the $\alpha\beta$ -heterodimer of TCRs the primary mediator of T cell-initiated adaptive immune responses.

The two-chain TCR $\alpha\beta$ -heterodimer is composed of an α - and a β -chain that dimerise to assume an immunoglobulin (Ig) fold [Figure 1]. The antigen specificity of the heterodimer is determined by the three complementarity-determining regions (CDRs) located at the TCR-pMHC-interface, and in particular the CDR3s, in the variable domains ($V\alpha$, $V\beta$) of either chain [5].

α - and β -chain variable domains are generated by VJ and VDJ recombination, respectively, from a genetically available catalogue of V-, D-, and J-genes [6]. While the CDR1 and CDR2 are entirely dictated by V-genes, the highly variable V(D)J recombination occurs in the CDR3 locus, making it the primary point of variation.

It is the independent and highly variable generation of the α - and β -chain by V(D)J recombination and the subsequent random pairing of chains that combinatorially establish the high-diversity TCR space needed to provide sufficient antigenic coverage [2, 7]. The T cell repertoire of an individual, i.e. the set of TCRs expressed across an individual's T cells, does not, however, encompass the full theoretical TCR space [8].

In practise, TCR repertoires are shaped both by constraints imposed during T cell maturation and by exposure of the adaptive immune system to antigens encountered throughout an individual's lifespan.

During T cell maturation, both recombined chains must first pass a series of selection steps that filter out non-functional chains. For instance, a significant portion of V(D)J recombinations result in out-of-frame se-

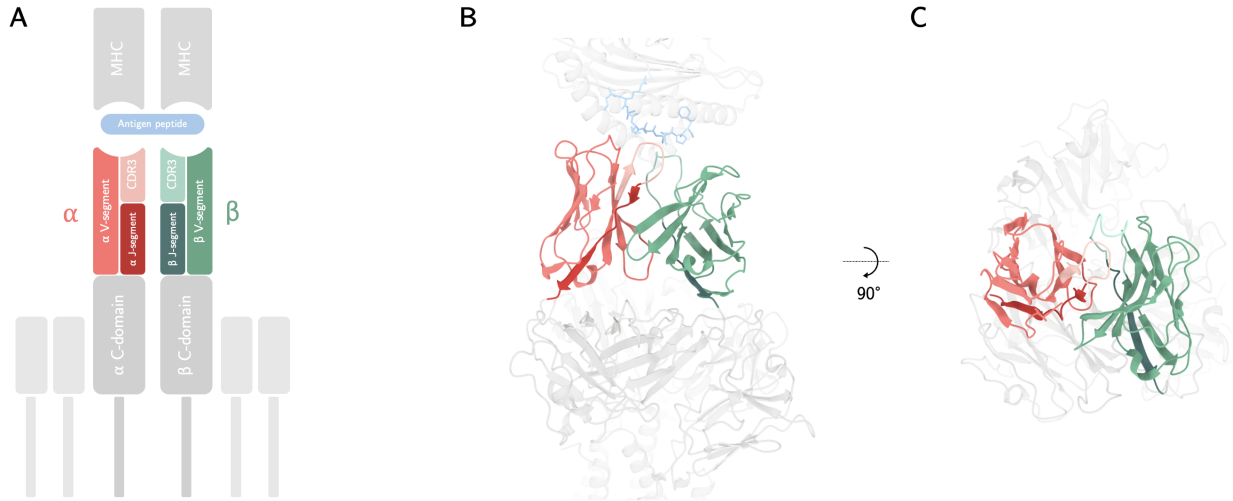


Figure 1: The TCR $\alpha\beta$ -heterodimer **A** Schematic of a TCR-pMHC interaction, with the α - and β -chains coloured by their gene segment origin. **B,C** An example protein structure coloured according to the schematic from a side and top-down view, respectively.

quences or premature stop codons, which are eliminated via nonsense-mediated decay or failed protein folding [9].

As α - and β -chain sequences are generated independently in each progenitor T cell, they are initially randomly paired. This pairing, however, must also pass a series of both developmental and antigen-driven selection steps [10]. The initial filter for an α - β -pairing is the physical ability of the independently rearranged chains to fold and dimerise within the endoplasmic reticulum. While the constant domains ($C\alpha$, $C\beta$) provide a structural scaffold, the variable domains ($V\alpha$, $V\beta$) must achieve proper inter-chain packing to ensure stable surface expression. Following the assembly of the $\alpha\beta$ -heterodimer and its translocation to the cell surface, nascent T cells undergo a two-step selection process within the thymus that assures T cells are both MHC-restricted and self-tolerant [11].

When exposed to their cognate antigens, mature T cells, and by extension their TCRs, undergo clonal expansion, increasing their population size [12]. On the other hand, T-cells that never engage with an antigen may become extinct due to disuse [13]. This selective expansion results in a non-uniform population distribution within TCR repertoires that favours both contemporarily and historically 'useful' TCRs.

By virtue of their high specificity, TCRs, and by extension their α - β -chain pairs, have a near one-to-one mapping to antigen epitopes. Consequently, TCR repertoires may be interpreted as bags of epitope labels that carry information about an individual's immune state and history. This has made TCR repertoires compelling subjects for immunological machine learning tasks, such

as epitope prediction and immune state classification [14, 15].

Immune repertoires are exceedingly diverse, estimated to contain on the order of 10^7 to 10^8 distinct TCRs [8, 16]. This diversity makes complete sampling of TCR immune repertoires challenging. As paired-chain $\alpha\beta$ sequencing is not yet feasible at the same high-throughput scale as single-chain sequencing [17, 18], many studies sequence only one of the TCR chain genes. Typically, the β -chain is sequenced due to its purported higher genetic diversity than the α -chain, arising from VDJ-recombination [19, 20]. Consequently, most attempts at building repertoire-level models train exclusively on TCR β -chain sequences [14, 15, 21–25].

However, to fully characterise a TCR, both the α - and β -chain encoding genes of a T cell should be sequenced together. The α -chain is known to have consequential influence on epitope specificity and affinity, and thus, on TCR function, with recent evidence suggesting that the paratope contribution of the α -chain is similar to that of the β -chain [20, 26]. Further, it has been found that α - and β -chains reach comparable structural diversity [20]. Thus, observing the β -chain in isolation may be insufficient for distinguishing TCR function.

Many machine learning projects largely proceed on the implicit assumption that β -chains are sufficient to distinguish the TCR epitope information necessary for repertoire-wide immune classification [10, 14, 15, 25]. The accuracy of this assumption has substantial implications for both diagnostic immunological sequencing and statistical and machine learning models that rely on repertoire data. If it does not hold, the cheaper, higher-throughput, more accessible single-chain sequencing ap-

proach may be less viable than originally assumed, and efforts should be concentrated on scaling and making the most of paired-chain sequencing approaches.

The assumption, however, may hold some merit. While any two α - and β -chains could theoretically pair, only a subset of pairs actually pass the selection criteria inherent to T cell maturation. Additionally, since the immune system is exposed to a finite set of antigens, only a smaller subset of TCRs will be engaged and persist in practice. Thus, it is not unreasonable to speculate that the combinatorially immense theoretical TCR space is only realised as a small subset of α - β -pairs so sparse that each unique β -chain is only ever observed to pair with one α -chain, and vice versa [7]. In such a case, when one of the two chains is observed, no new information would be gained by observing its partner chain; it is, so to speak, deterministic.

Even if partnering diversity is observed, it may still be inconsequential in distinguishing the TCR epitope. If, for instance, a β -chain is observed to pair with several α -chains, but all are nearly sequence- and structure-wise identical, odds may be that any pairing will engage the same epitope.

In other words, if neither differential pairing nor functional divergence between their α -partners is observed for β -chains, β -chains may serve as coarse-grained functional proxies for $\alpha\beta$ -heterodimers. This two-dimensional diversity concept is illustrated in [Figure 2B].

In this study, the *Observed T cell receptor Space* (OTS) database [18], the largest public paired-chain TCR database, is utilised to examine whether such diversity is realised in TCR repertoire samples, by examining the extent to which β -chains uniquely identify their α -chain partner, and how functionally distinct these α -chain partners are.

Establishing this is challenging since existing paired-chain TCR datasets are profoundly undersampled relative to the expected diversity of TCR repertoires, due to the limitations of paired-chain sequencing [10]. Therefore, analyses will include sampling depth of the TCRs as a covariate, under the assumption that more richly sampled TCRs will yield a better estimate of the underlying distribution of diversity.

The primary objective of this study is not to establish converged estimates of certain diversity measures, but rather to demonstrate that pairing diversity may be non-zero and functionally consequential, and therefore non-negligible in single-chain TCR repertoires. As the OTS sample size grows over time, the finite sampling and undersampling biases that may dominate analyses in the OTS now, should diminish. Highly sampled sets of TCRs thus come to represent a forecast of what we expect to observe in the future OTS. While many of

these analyses may yield more robust results if rerun in the future, where data support is greater, establishing an impression of the regime of TCR pairing diversity we face now is of consequential importance to current efforts to develop diagnostic models that rely on TCR repertoires.

2 Methods

2.1 Curation of paired TCR sequences

Paired chain sequences were sourced from the *Observed T cell receptor Space* (OTS) database [18].

The OTS was filtered to human, non-gamma-delta T cell-type data units, and to sequences where all amino acid sequence and gene call information was available for both chains. This yielded 5,083,688 paired chain entries across 2,129 data units from 62 studies.

Data units were aggregated into repertoires if they originated from the same Study and Subject. If no Subject information was available, each data unit was assumed to derive from a distinct subject. This aggregation pools sequences from different tissue samples and different T cell types and subtypes. This yielded 1,733 repertoires. Repertoires are characterised in [Figure S1].

Unique α and β -chains were defined in two ways: Either by V-gene, J-gene, and CDR3 sequence identity (*Gene definition*), or by full amino acid sequence identity from FRW1 to FRW4 (*Sequence definition*). The *gene definition* approach yields 1,544,941 unique pairs from 1,020,192 unique α - and 1,473,969 unique β -chains. The *sequence definition* yields 1,552,458 unique pairs from 1,026,738 unique α - and 1,464,973 unique β -chains. Analyses were found to be invariant to using either definition (As an example, see [Figure S2 and S3]), so the *gene definition* is used throughout.

apply the *gene chain definition* unless otherwise specified.

Analyses where all paired chain entries from OTS data units are pooled together are referred to as *pan-repertoire*. In contrast, contexts where analyses are performed within or between repertoires are referred to as *intra-* or *inter-repertoire*, respectively. For short, these contexts are denoted *pan*, *intra*, and *inter*.

2.2 Observation measures

The following analyses will utilise three distinct measures of how often chains or pairs of chains are observed.

Instances N measures how many times a chain or pair is observed as an entry, i.e. as a row. In contrast, **repertoire occurrences** N_r measures how many distinct repertoires each chain or pair is observed in.

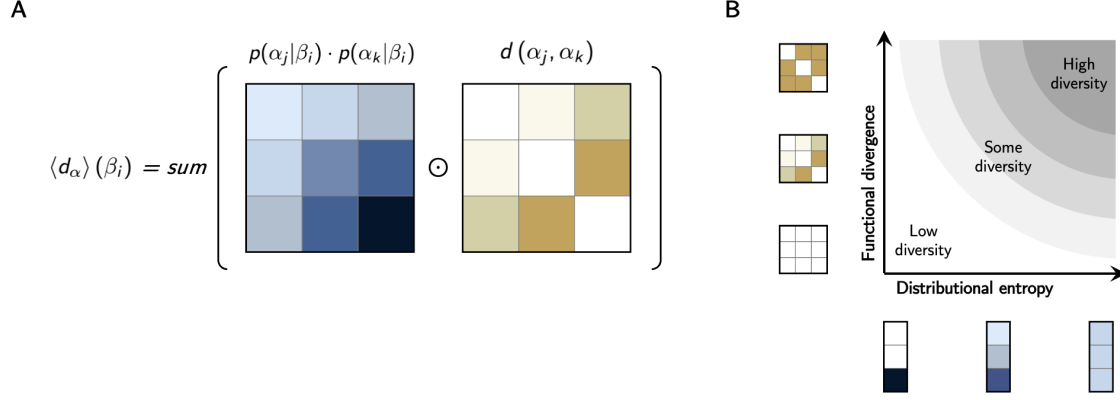


Figure 2: The concept of 'pairing diversity' has both a distributional and a functional aspect. **A** Conceptualisation of the expected distance $\langle d_\alpha \rangle(\beta)$ measure (see Methods 2.3.4). Each row and each column represents one α -chain, such that each cell represents one pair of α -chains. The blue matrix represents the probability that two chains are co-sampled in a sample of two TCRs with the same β -chain. The yellow matrix represents some functional distance between the two α -chains. The sum over their element-wise product becomes a weighted average of the functional distance between α -chains. **B** Biologically significant pairing diversity only emerges if different chain pairings are both likely to occur and functionally distinct. I.e. only if probability mass is distributed amongst different chain pairing outcomes (distributional entropy), and if there are some non-zero functional differences between pairing outcomes (functional divergence), such as differing epitope specificities.

The shallow sampling depth of TCRs in the OTS warants both of these observation measures. On one hand, more instances N of a TCR within repertoires not only lend statistical support but also contain valuable information about the entropy of pairing. On the other hand, repertoires in the OTS vary significantly in sample size [Figure S1A,B], which may bias towards certain repertoires. Repertoire occurrence N_r is agnostic to repertoire sample size, alleviating this bias.

Frequency p of a specific chain or pair is defined as its relative number of instances N in a given set.

2.3 Diversity measures

Several measures are introduced to probe the differential pairing of α - and β -chains. As most single-chain TCR repertoires utilised in research today are β -repertoires, analyses will be performed in the context of discerning distinct α -pairings given a specific β -chain, $\alpha_j|\beta_i$. The following measures are thus all derived from the set of α -chains $\{\alpha\}_i$ that are observed to pair with a specific β -chain β_i at frequency $p(\alpha_j|\beta_i)$.

2.3.1 Unique pairings $S_\alpha(\beta)$

The number of unique pairings $S_\alpha(\beta)$ naïvely counts the number of unique α -chains that a specific β -chain has been observed to pair with. It is defined as the length of the set of α -chain pairings

$$S_\alpha(\beta_i) = |\{\alpha\}_i| \quad (1)$$

2.3.2 Perplexity of pairing $\hat{S}_\alpha(\beta)$

The perplexity of pairing $\hat{S}_\alpha(\beta)$ takes into account the frequency at which each unique α -chains is observed to pair with a specific β -chain. It is defined as the perplexity over $p(\alpha_j|\beta_i)$

$$\hat{S}_\alpha(\beta_i) = 2^{H_\alpha(\beta_i)} \quad (2)$$

Where $H_\alpha(\beta_i)$ is the Shannon entropy over $p(\alpha_j|\beta_i)$

$$H_\alpha(\beta_i) = - \sum_{\alpha_j \in \{\alpha\}_i} p(\alpha_j|\beta_i) \cdot \log_2 p(\alpha_j|\beta_i) \quad (3)$$

It can be interpreted as the effective number of unique pairings.

2.3.3 Simpson index of pairing $\lambda_\alpha(\beta)$

The Simpson index of pairing $\lambda_\alpha(\beta)$ measures the probability that two randomly sampled α - β -pairs, $(\alpha_j, \beta_i), (\alpha_k, \beta_i)$ that share the same β -chain β_i also share the same α -chain $\alpha_j = \alpha_k$

$$\lambda_\alpha(\beta_i) = \sum_{\alpha_j \in \{\alpha\}_i} p(\alpha_j|\beta_i)^2 \quad (4)$$

In other words, if you sample the same β -chain twice, $\lambda_\alpha(\beta)$ indicates the probability that the two pairs are identical.

In learning tasks where a signal becomes associated with a specific β -chain in one repertoire, the Simpson

index $\lambda_\alpha(\beta)$ effectively measures the specificity with which you can extrapolate that signal to the same β -chain in other repertoires, assuming that the signal is also α -chain specific. Since several α -chains may fulfil the same function given the same β -partner, this interpretation is not rigorously precise, unless α -chains are grouped based on their shared function when paired with the specific β -chain.

2.3.4 Expected distance $\langle d_\alpha \rangle(\beta)$

The expected distance $\langle d_\alpha \rangle(\beta)$ is a weighted average of some chain distance measure d between all pairs in $\{\alpha\}_i \times \{\alpha\}_i$

$$\langle d_\alpha \rangle(\beta_i) = \sum_{\alpha_j \in \{\alpha\}_i} \sum_{\alpha_k \in \{\alpha\}_i} p(\alpha_j|\beta_i) \cdot p(\alpha_k|\beta_i) \cdot d(\alpha_j, \alpha_k) \quad (5)$$

It can be interpreted as the expected distance between α -chains from two samples that share a specific β -chain β_i . See [Figure 2A].

Several inter-chain distance measures can be utilised to capture the functional divergence of α -chains. Some of the more directly interpretable distance measures include the Levenshtein distance d_L between two α -chain sequences, which can be used to calculate the **expected Levenshtein distance** $\langle d_{L,\alpha} \rangle$.

Additionally, one can limit chain comparison to specific subfeatures, such as to compare chains by their **expected CDR3 Levenshtein distance** $\langle d_{L,\alpha\text{CDR3}} \rangle(\beta)$.

2.4 Structural modelling with TCR-Builder2

TCR protein structures were generated from α - and β -chain amino acid sequences with the TCRBuilder2 [20] model from the ImmuneBuilder suite (github.com/oxpig/ImmuneBuilder) using the TCR-Builder2+ weights.

Structures were visualised with ChimeraX [27] and analysed with MDAnalysis [28].

The example structures in [Figure 8] are generated with TCRBuilder2 using the chain amino acid sequences found in **supplementary sequences**.

3 Results

This study discerns TCR pairing diversity from the perspective of β -chains.

For a more nuanced characterisation of pairing diversity, analyses are performed across four measures that vary in interpretability, complexity, and context-richness: The naïve number of unique pairings $S_\alpha(\beta_i)$, the perplexity of pairing $\hat{S}_\alpha(\beta_i)$, the Simpson index of pairing

$\lambda_\alpha(\beta_i)$, and the expected Levenshtein distance between α -chain partners $\langle d_{L,\alpha} \rangle(\beta_i)$.

Contemporary paired-chain TCR repertoire datasets are severely undersampled relative to the expected diversity of the immune space. Most pairs are observed only once or a few times, making it difficult to distinguish biological diversity from sampling noise. Therefore, analyses are performed with sampling depth of the TCRs, either represented by the number of instances $N(\beta_i)$ or the number of repertoire occurrences $N_r(\beta_i)$ of chains, as a covariate. Undersampling is expected to downward-bias the entropy of the pairing distributions, and by extension, the diversity measures. As such, estimates of pairing diversity presented in this study may be more akin to a lower bound, and future analyses with larger datasets can be expected to yield higher diversity estimates than reported here.

Analyses will be performed both in *pan-repertoire* and *inter-repertoire* contexts. The pan-repertoire context, in which all repertoires are pooled, yields the largest data support and likely represents the best estimate of a true common TCR distribution, if such a distribution exists. It is, however, challenged by imbalanced repertoire sizes. Additionally, germline genetics, thymic selection history, and shared disease exposure differences may upward-bias pairing diversity in a way that does not accurately reflect true chain pairing promiscuity. On the other hand, individual repertoires suffer from individual biases, such as disease states, and shallow sampling depths, which intensify finite-sample effects in diversity measures. By contrasting pan-repertoire and intra-repertoire analyses, one can better discern whether observed diversity reflects genuine biological diversity or cross-individual sharing of public TCRs.

3.1 Public paired chain TCRs are observed frequently in the OTS

Most TCR chains in the OTS are observed very few times. α -chain, β -chain, and α - β -pair observations are distributed according to a power law, whether measured by the number of instances N or repertoire occurrences N_r [Figure S2A-D Top]. This is expected in sparse sampling of a highly diverse space and is consistent with previous findings in single-chain repertoires [8, 29].

Despite this, a significant portion of the TCRs in the OTS are observed inter-repertoire, i.e. across more than one repertoire sample [Figure S2A-D Bottom]. Such chains and pairs observed across many individuals are known as 'public' TCRs.

While there is no consensus threshold for how large a proportion of repertoires a TCR should be observed in to be considered public, the greedy approach is to include all TCRs that appear in at least two repertoires. [Figure

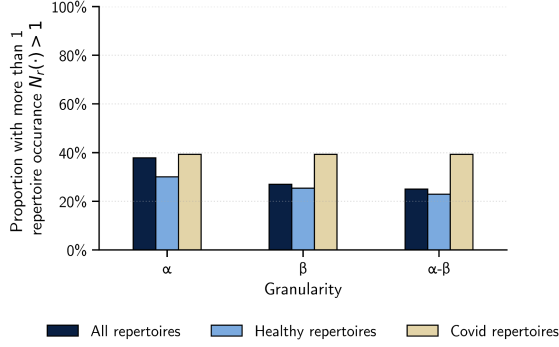


Figure 3: The proportion of α -chains, β -chains, or α - β -chain pairs that are observed across multiple repertoires. The dark blue bars are calculated using all repertoires, whereas the light blue and light yellow bars are calculated using subsets of repertoires filtered by their disease annotation being either healthy or covid (See [Figure S1C]).

3] shows that about 20% of TCR α - β -pairs in the OTS are observed in more than one repertoire, about 30% of β -chains, and about 40% of α -chains. Notably, these inter-repertoire rates are much higher than corresponding findings for B-cell receptors [30], which undergo somatic hypermutation. This demonstrates the suitability of TCRs for extrapolating learnings between repertoires.

Repertoires sharing pathogenic disease labels are expected to have undergone convergent immune responses to a shared antigen. Not only will these convergent TCRs be shared across repertoires, but they will likely also be clonally expanded as a consequence of cognate antigen exposure, making them more likely to be sampled than other TCRs that do not recognise the shared antigen. This is expected to upward-bias inter-repertoire observations of disease-related TCRs, as can be observed in the yellow bars of [Figure 3] that are based on the subset of the OTS with the covid disease label.

Since the bulk of the OTS originates from repertoires with the covid disease label [Figure S1C], these convergent immune repertoires are expected to upward-bias the proportion of TCRs observed in more than one repertoire. For contrast, this proportion is also plotted for the subset of the OTS with the healthy label, which exhibits lower proportions of inter-repertoire TCRs. This may also be explained, however, by the fact that the healthy subset is smaller than both the full OTS and the covid subset, and that undersampling can make shared TCRs appear individual-specific.

3.2 β -chains are observed to pair with multiple α -chains both across and within repertoires

Most β -chains in the OTS are only observed to pair with one α -chain [Table 1], specifically about 96%. This low proportion of observed differential pairing is largely due to many OTS entries being sampled only once. The following sections, however, will take sampling depth into account.

Despite this, many β -chains are nonetheless observed to pair with two, three, or a multitude of distinct α -chains. Critically, this finding holds intra-repertoire. In other words, even within the same repertoire, β -chains can be observed with multiple α -partners. This diversity cannot be explained by inter-individual germline variation or exposure history, and must reflect genuine α - β -pairing multiplicity.

These findings extend to the opposite perspective; α -chains are conversely observed to pair with multiple β -chains.

3.3 β -chains are increasingly observed to pair with multiple α -chains with increasing repertoire sampling

Sampling the full TCR repertoire of an immune system is exceedingly difficult. It is challenged by a series of hurdles, including the spatial distribution of T-cells across tissue types, the methodological constraints imposed on sequencing leukocytes from blood samples, the throughput bottleneck of paired-chain sequencing, the large diversity of the TCR space, and the non-uniform distribution of TCR abundances. Achieving an exhaustive, converged TCR repertoire is, and will likely remain, a major experimental challenge.

As sampling depth increases, TCR repertoires are expected to converge to their true distribution. As the distribution settles, so ideally would α - β -pairing diversity measures too.

In the limit of exhaustive sampling, the number of unique α -pairings $S_\alpha(\beta_i)$ given a β -chain β_i should approach the (likely very large) number of surface expressible α - β -pairs that can pass thymic selection. As this number is expected to be exceedingly large and exhaustive sampling is far beyond reach, $S_\alpha(\beta_i)$ is expected to continually increase with sampling depth.

When observing $S_\alpha(\beta_i)$ as a function of the number of observed instances $N(\beta_i)$ of a β -chain, $S_\alpha(\beta_i)$ indeed seems to increase [Figure 4A], at least when considering the median and mean. When observing $S_\alpha(\beta_i)$ as a function of the number of repertoire occurrences $N_r(\beta_i)$ of a β -chain, the continuous increase of $S_\alpha(\beta_i)$ is much more apparent [Figure 4B].

Unique pairings S	Pan-repertoire		Intra-repertoire	
	α -chains	β -chains	α -chains	β -chains
1	833,139	1,418,999	3,051,853	3,122,222
2	98,898	45,865	41,770	29,960
3	35,345	6,134	5,511	2,485
4	17,500	1,664	1,833	495
5+	35,310	1,307	3,128	273
Total	1,020,192	1,473,969	3,104,095	3,155,435

Table 1: Unique chain pairings $S_\beta(\alpha_j)$, $S_\alpha(\beta_i)$ given either the α - or the β -chain, both in the *pan*- and *intra-repertoire* context. I.e., 6,134 β -chains are observed to pair with 2 distinct α -chains *pan-repertoire*. Since the same chain may be present in multiple repertoires, the total amount of chains in the *intra-repertoire* context appears higher than the total amount of unique chains.

In the context of statistical and machine learning models based on TCR repertoires, TCRs observed across repertoires are of special interest, as they are well-suited for inter-repertoire extrapolation of signals.

Interestingly, even in cases where β -chains have only been sampled in very few instances, some of these are still observed to pair with many (>5) α -chains.

In the *intra-repertoire* context, the expected increase of $S_\alpha(\beta_i)$ is somewhat observed [Figure 5A]. In any case, even *intra-repertoire* a substantial amount of unique α -chain pairing is observed given enough sampling depth.

3.4 The perplexity of α -pairings given a β -chain indicates a more modest degree of pairing diversity

While the number of observed chain pairings $S_\alpha(\beta_i)$ demonstrates that differential pairing is occurring, it does not capture the effective diversity well. A pairing observed 10,000 times contributes just as much to $S_\alpha(\beta_i)$ as a pairing observed just once.

Pairing perplexity $\hat{S}_\alpha(\beta_i)$, derived from the entropy over the distribution of observed α -pairings given a β -chain $\alpha|\beta_i$, provides a measure of the effective number of unique pairings observed. Say a β -chain β_i is observed to pair with one α -chain α_j 99.9% of the time and 10 other α -chains the rest of the time, then $\hat{S}_\alpha(\beta_i) \approx 1$ whereas $S_\alpha(\beta_i) = 11$. Clearly, the perplexity of pairing provides a more nuanced take on pairing diversity.

In the limit of exhaustive sampling, a small set of α -chain partners that enable the specific binding to some probable antigen would be expected to dominate as partners for a specific β -chain. Thus, pairing perplexity is expected to converge to a specific value for each β -chain. By extension, the distribution of pairing perplexity estimates would also be expected to converge.

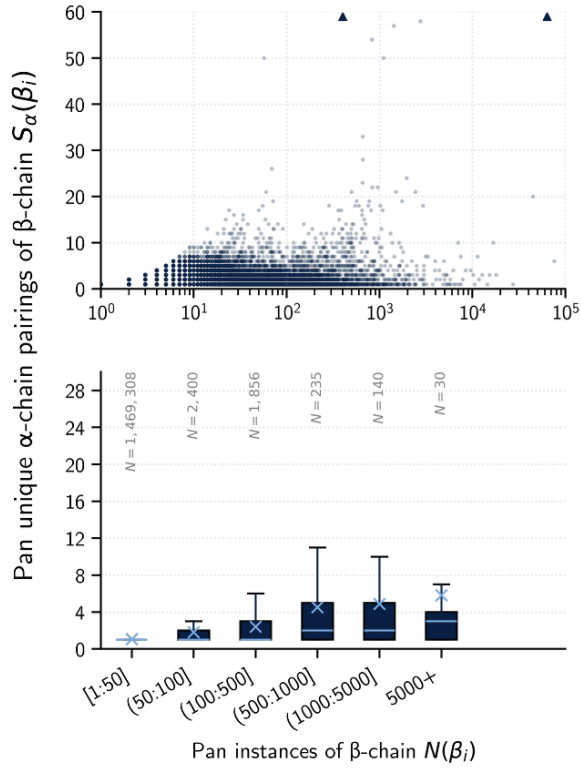
When observing $\hat{S}_\alpha(\beta_i)$ as a function of the number of observed instances $N(\beta_i)$ of a β -chain, $\hat{S}_\alpha(\beta_i)$ appears to primarily converge to 1 [Figure 4C]. The same behaviour is observed *intra-repertoire* [Figure 5B].

When observing $\hat{S}_\alpha(\beta_i)$ as a function of the number of repertoire occurrences $N_r(\beta_i)$ of a β -chain, however, a distribution over $\hat{S}_\alpha(\beta_i)$ appears to converge [Figure 4D]. This distribution appears consistent across varying sampling depths, starting at about 50 repertoire occurrences.

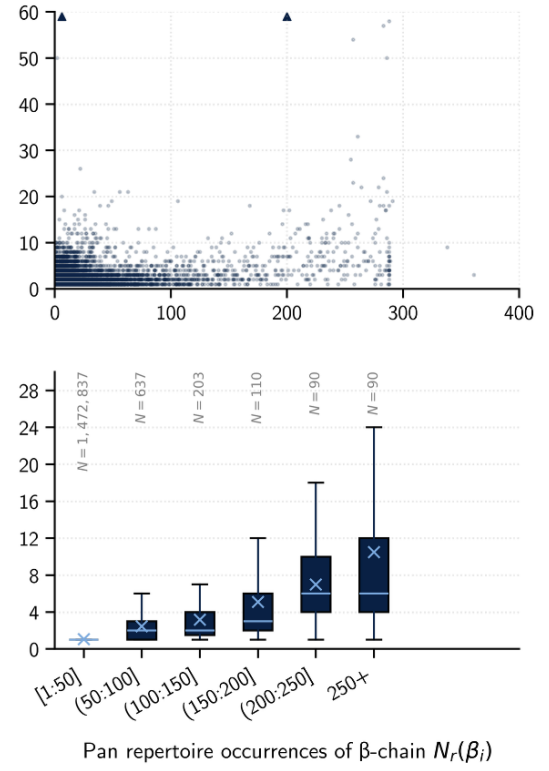
Generally, when considering either total *pan-repertoire* instances N or repertoire occurrences N_r for determining when chains are in a high-sampling regime, analyses can arrive at somewhat different conclusions about convergence. While both observation measures have their merits, instances N are likely heavily affected by sampling artefacts, such as some pairs being highly sequenced within just a single repertoire, inflating N without adding additional inter-repertoire evidence. Repertoire occurrences N_r , however, provide a reliable signal of genuine inter-repertoire distributional entropy. Additionally, analyses utilising repertoire occurrences as the sampling depth measure often show more converged diversity distributions that match expectations, whereas instance-based analyses are often highly heteroscedastic. Thus, findings based on sufficient repertoire occurrence N_r sampling are likely the most reliable.

These findings, in any case, indicate that the number of unique pairings $S_\alpha(\beta_i)$ is indeed severely overestimating the true biological diversity of α - β -pairing. The distribution of the frequency with which a β -chain pairs with each of its α -chain partners is highly skewed; It appears that, generally, one or two α -chain pairings dominate. However, observing a distribution also demonstrates that pairing promiscuity is indeed chain-dependent, and that the α -chain partner given a β -chain cannot be deterministically inferred *a priori*. Sometimes there is effectively only one possible pairing, some-

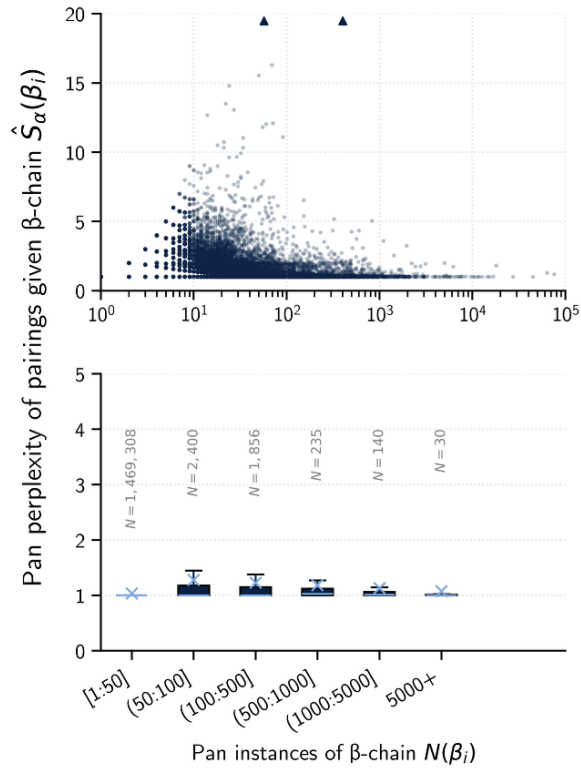
A



B



C



D

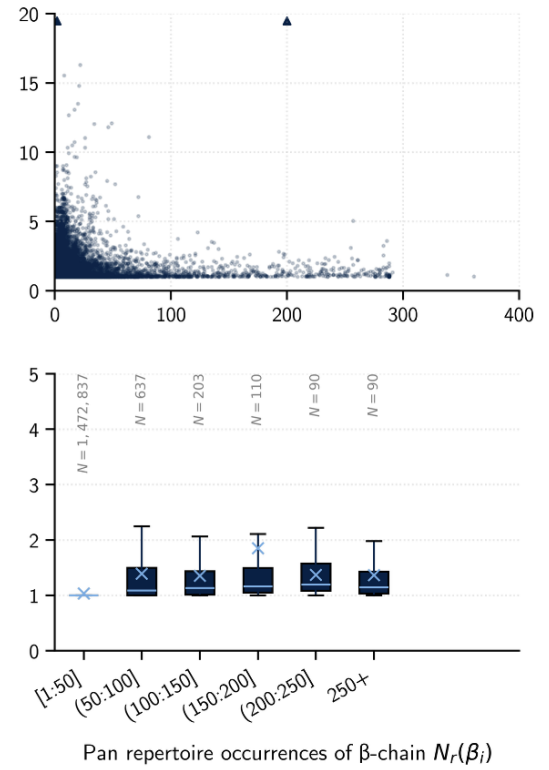


Figure 4: [Previous page] Pan-repertoire α -chain pairings per β -chain $S_\alpha(\beta_i)$. **A Top** Scatter plot, where each β -chain β_i is plotted by the its number of *pan-repertoire* instances in the OTS $N(\beta_i)$ and by how many *pan-repertoire* unique α -chains it is observed to pair with $S_\alpha(\beta_i)$. **Bottom** Box plots of the same data with outliers outside the IQR omitted. Light blue lines and crosses are medians and means, respectively. Grey numbers above box plots indicate the number of observations in each column. **B** The same as **A**, but where each β -chain β_i is plotted by the the number of *pan-repertoire* repertoire occurrences in the OTS $N_r(\beta_i)$ instead of by its instances. **C,D** are the same as **A,B**, but where each β -chain β_i is plotted by its *pan-repertoire* perplexity of pairing $\hat{S}_\alpha(\beta_i)$ instead of by its unique pairings. While the two measures are comparable in interpretation (**A** and **B** show the *observed* number of pairs, whereas **C** and **D** show the *effective* number of pairs), please note the difference in scale of the Y-axes.

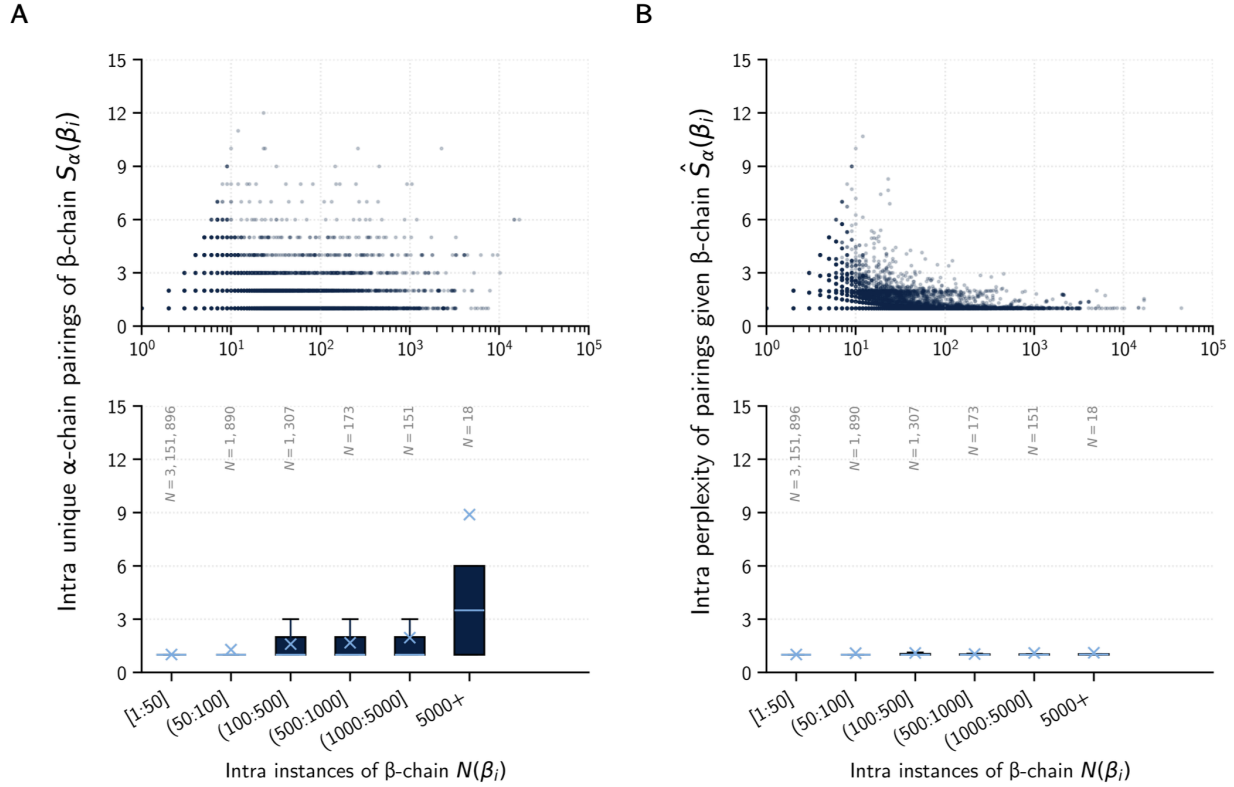


Figure 5: Intra-repertoire α -chain pairings per β -chain $S_\alpha(\beta_i)$. **A Top** Scatter plot, where each β -chain β_i is plotted by the its number of *intra-repertoire* instances in the OTS $N(\beta_i)$ and by how many *intra-repertoire* unique α -chains it is observed to pair with $S_\alpha(\beta_i)$. Triangles indicate out-of-axis-limits points. **Bottom** Box plots of the same data with outliers outside the IQR omitted. Light blue lines and crosses are medians and means, respectively. Grey numbers above box plots indicate the number of observations in each column. **B** The same as **A**, but where each β -chain β_i is plotted by its *intra-repertoire* perplexity of pairing $\hat{S}_\alpha(\beta_i)$ instead of by its unique pairings. This figure can be interpreted as an *intra-repertoire* version of [Figure 4A,C].

times there are multiple. These findings would indicate that there is genuine distributional entropy in TCR α - β -pairing.

Interestingly, while $\hat{S}_\alpha(\beta_i)$ estimates are generally much lower than $S_\alpha(\beta_i)$, there are still many β -chains that do show pairing perplexities of 5 and above, even for highly-sampled β -chains.

The Simpson index of pairing $\lambda_\alpha(\beta_i)$ also takes into account the frequency with which a specific β -chain pairs with its α -chain partners. It represents the probability that two randomly sampled TCRs that share the same β -chain also share the same α -chain. A value of 1 indicates complete certainty that the α -chain is the same in both samples, whereas a value near 0 indicates that it would be highly likely that the α -chains are distinct. The interpretability of this measure makes it well-suited for evaluating the specificity with which learnt signals from a β -chain sampled in one repertoire can be extrapolated to the same β -chain in another repertoire.

When observing $\lambda_\alpha(\beta_i)$ as a function of the number of observed instances $N(\beta_i)$ of a β -chain, $\lambda_\alpha(\beta_i)$ appears to converge to 1, whether *pan*- or *intra*-repertoire [Figure S4A,C].

When observing $\lambda_\alpha(\beta_i)$ as a function of the number of repertoire occurrences $N_r(\beta_i)$ of a β -chain, however, a distribution over $\lambda_\alpha(\beta_i)$ appears to converge [Figure S4B]. This distribution appears consistent across varying sampling depths, starting at about 50 repertoire occurrences. It is unclear, however, whether the distribution narrows as sampling depth increases.

The Simpson index recapitulates many of the findings of the perplexity of pairing, and also demonstrates the presence of non-negligible chain pairing in learning tasks.

3.5 α -chains that pair with the same β -chain can exhibit significant sequence differences

Distributional pairing diversity on its own is not sufficient to conclude that β -chains in isolation are insufficient to deduce TCR function. While the α -chain partner also contributes to TCR specificity, the β -chain may, in theory, bind the same antigen regardless of which of its common α -chain partners it is observed with, if all α -chains are functionally similar.

As no well-generalising high-throughput TCR epitope prediction model has yet been published, other proxies and tools must be utilised to discern whether distinct α -chains will result in different TCR epitope specificities when paired with a specific β -chain.

One such functional proxy is the Levenshtein distance d_L , colloquially 'edit distance', which counts the number of amino acid substitutions and/or indels required

to transform one sequence into another. While function may be conserved for some substitutions (such as Asp to Glu), and while some substitutions or indels might be more impactful than others, Levenshtein distance nonetheless represents an interpretable measure that could indicate potential functional divergence.

The pairwise α -chain Levenshtein sequence distance will be evaluated over the α -chain pairing frequencies given a β -chain $p(\alpha|\beta_i)$ as an expected distance measure $\langle d_{L,\alpha} \rangle$ [Figure 2]. The expected Levenshtein distance can be interpreted as the Levenshtein distance one would expect between the α -chains from two samples that share a specific β -chain β_i .

In the simplest non-trivial case, a β -chain is observed to pair with two α -chains. If one chain dominates, the effective distance will be $\langle d_{L,\alpha} \rangle \approx 0$. If either α -chain is equally likely to be the partner, i.e. a maximum entropy scenario, the effective distance will be half the Levenshtein distance between them.

The Levenshtein distance will be evaluated between all three of the α CDRs, $\langle d_{L,\alpha CDRs} \rangle$, which all contribute to antigen specificity. As the CDR3 is known to be the largest contributor to both diversity and specificity, the Levenshtein distance will also be evaluated between just the α CDR3s $\langle d_{L,\alpha CDR3} \rangle$.

When observing $\langle d_{L,\alpha CDRs} \rangle$ as a function of the number of observed instances $N(\beta_i)$ of a β -chain, $\langle d_{L,\alpha CDRs} \rangle$ appears to converge to between 0 and 1 [Figure S6A].

When observing $\langle d_{L,\alpha CDRs} \rangle$ as a function of the number of repertoire occurrences $N_r(\beta_i)$ of a β -chain, once again, a distribution over $\langle d_{L,\alpha CDRs} \rangle$ appears to converge [Figure 6B]. This distribution appears consistent across varying sampling depths, starting at about 50 repertoire occurrences.

Limiting the scope to the CDR3 with $\langle d_{L,\alpha CDR3} \rangle$ [Figure 6C,D] recapitulates the behaviour found for $\langle d_{L,\alpha CDRs} \rangle$, and reveals that the CDR3 contributes about half of the expected Levenshtein distance of $\langle d_{L,\alpha CDRs} \rangle$.

Interestingly, the corresponding intra-repertoire analyses [Figure 7] determine much the same range of expected α -chain Levenshtein distance values. I.e., large expected α CDR Levenshtein distances can be expected even within repertoires.

It is practical to interpret the expected CDR Levenshtein distances in the context of typical CDR lengths [Figure S5], which are about 5-7 amino acids, 5-8 amino acids, and 9-16 amino acids, respectively, for CDR1, -2, and -3. The combined CDR length is typically about 21-29 amino acids.

While the mean and median expected distance are relatively low in the lens of typical CDR lengths, many of the observed high-value expected Levenshtein distances indicate sequence identities below 50%. While Leven-

shstein distance is not a perfect proxy for conserved function, this range of low sequence identity is often quite consequential.

Overall, these findings indicate that while low-diversity β -chain species dominate, distributionally diverse and functionally divergent α -chain pairings do exist for a significant portion of observed β -chains. α -chain pairings may differ substantially in the regions determining antigen specificity.

With both distributional entropy, as represented by perplexity or the Simpson index of pairing, and functional divergence, represented by the expected Levenshtein distance, a map akin to the concept in [Figure 2B] has been realised with data in [Figure S6] to illustrate the different domains of chain pairing diversity.

This figure demonstrates that a range of β -chains exist, which all have very low Simpson indices of pairing $\lambda_{\alpha}(\beta_i)$, but highly variable expected sequence differences between α CDRs. This may indicate two distinct regimes of multi- α pairing β -chains: Some β -chains (in the lower right corner of the plot) that consistently attract distinct but functionally similar α -chains (low $\langle d_{L,\alpha CDRs} \rangle$), and other β -chains (in the upper right corner) that pair with structurally diverse α -chains (high $\langle d_{L,\alpha CDRs} \rangle$).

The first of these regimes (low $\langle d_{L,\alpha CDRs} \rangle$) may reflect β -chains which are strongly dependent on the right α -chain partner for binding a specific epitope.

The second of these regimes (high $\langle d_{L,\alpha CDRs} \rangle$) could perhaps reflect 'dominant' β -chains, whose epitope specificity is α -chain agnostic, or perhaps 'universal' β -chains, which can engage a range of different epitopes depending on their α -partner.

3.6 α -chains that pair with the same β -chain can exhibit significant structural differences

While functional proxies like the Levenshtein distance are useful for bulk assessment of TCRs, true divergence in binding specificity is much more distinguishable from structural differences.

As a proof-of-concept of observable structural diversity for distinct α -chains observed to pair with the same β -chain, a highly-sampled β -chain has been selected as an example that demonstrates both high pairing diversity and expected α CDR-distance.

[Figure 8] shows three TCR structures generated by TCRBuilder2 from three α - β -pairs sharing the same β -chain β_i . There are 43 instances of β_i pairing with α_1 across 5 repertoires, 13,213 instances of pairing with α_2 across 42 repertoires, and 4,815 instances of pairing with α_3 across 36 repertoires in the OTS. The

sequences for the three chains [Supplementary sequences] show that while α_1 and α_2 only differ by one arginine-methionine substitution in their CDR3, α_3 is an entirely different α -chain.

This is apparent from the structures in [Figure 8] that clearly show a very different α CDR3 loop conformation for α_3 . Besides the highly diverging backbone conformations shown in [Figure 8], electrostatic and hydrophobic surface maps generated with ChimeraX exhibited noticeably different CDR binding interface profiles - even between α_1 and α_2 , as the arginine introduces a positively charged potential in the interface.

Notably, α_2 and α_3 are both observed to pair with β_i frequently, yet exhibit wholly different $\alpha\beta$ -structures, and presumably engage different antigens.

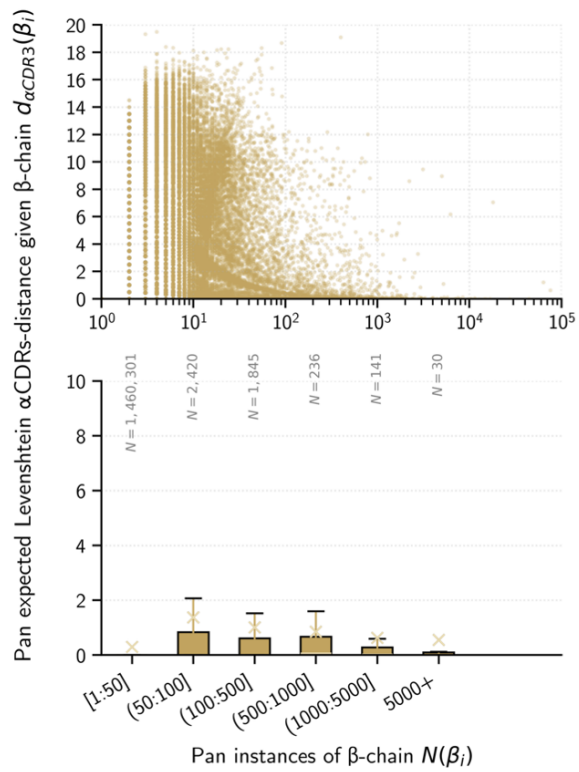
4 Discussion

Single-chain β TCR repertoires represent large and information-rich datasets for immunoinformatic learning tasks. As such, TCR repertoire sequencing is increasingly being employed as a diagnostic readout to train machine learning models to classify immune states, infection history, or autoimmunogenic disorders. Most of this work relies on single-chain β repertoires. These repertoires, however, may be fundamentally limited in their ability to extrapolate learned signals between repertoires, due to the inherent promiscuity of the unobserved, but functionally consequential, α -chain partner.

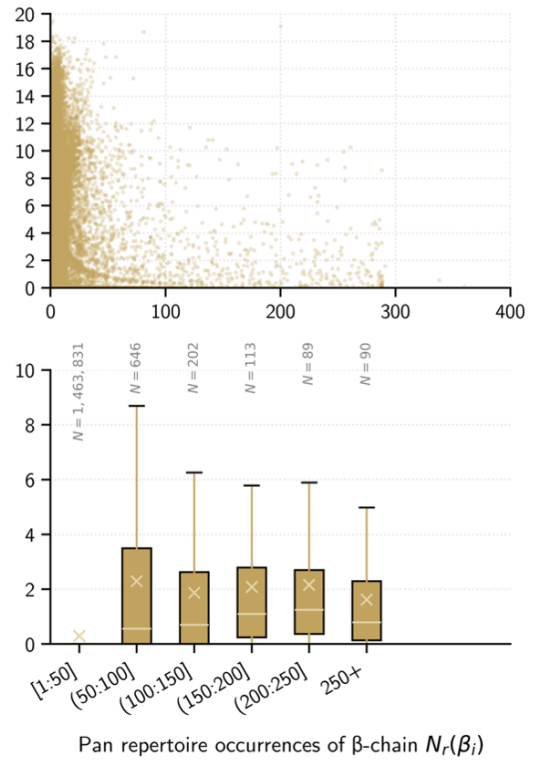
This study set out to quantify the extent to which β -chains are observed to pair with multiple α -chain partners in the OTS, the largest database of paired-chain TCR sequences. It was found that while contemporary paired chain repertoire datasets are severely undersampled, and while the majority of β -chains are only observed with one α -chain partner, there is still an abundance of evidence that β -chains may effectively pair with multiple distinct α -chain partners, even within repertoires, and even when accounting for the relative frequencies with which each pairing was observed. Whether there is some way to infer, *a posteriori*, which β -chains are α -chain promiscuous, so to speak, remains an open question. Until such classification of β -chains is possible, however, the demonstrated α -chain pairing diversity of β -chains could equally apply to any newly observed β -chain, and ought to be taken into account when designing models based on single-chain β repertoires.

To establish whether such pairing diversity is consequential for TCR epitope specificity, the primary feature setting TCRs apart, functional divergence of pairings was evaluated using the expected Levenshtein distances

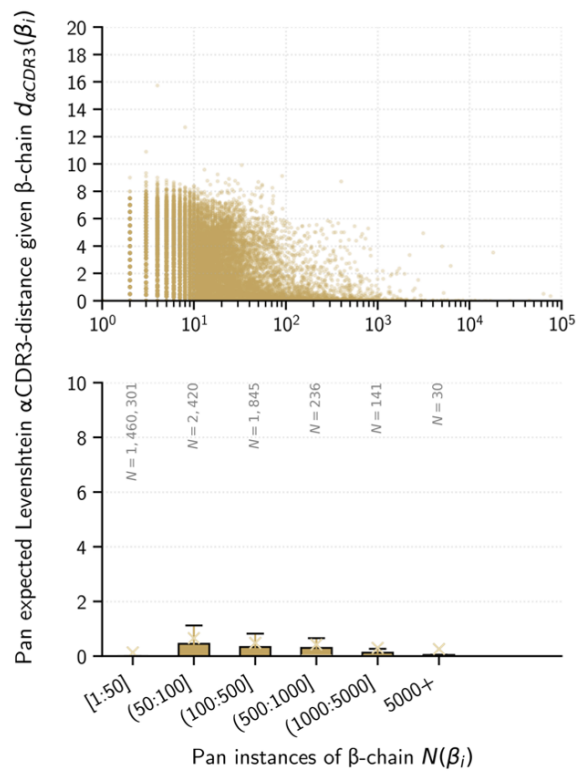
A



B



C



D

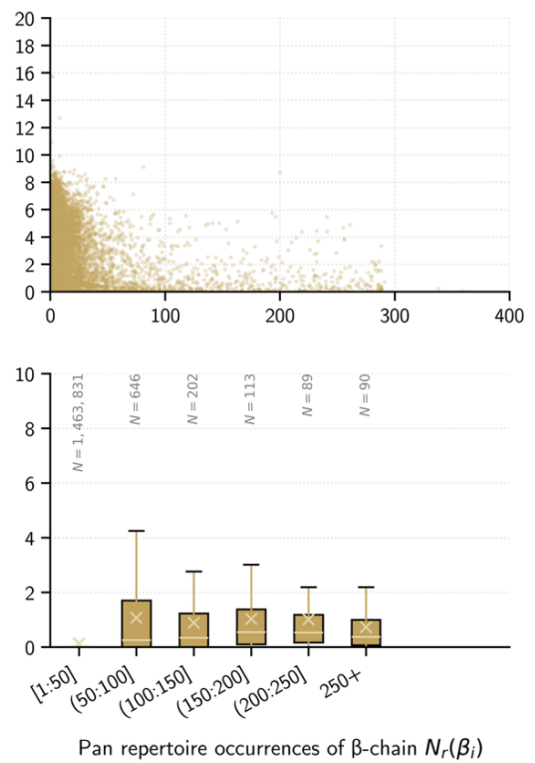


Figure 6: [Previous page] Pan-repertoire expected Levenshtein distance between α -chains given the β -chain. **A Top** Scatter plot, where each β -chain β_i is plotted by β_i is plotted by the its number of *pan-repertoire* instances in the OTS $N(\beta_i)$ and by the expected Levenshtein distance between the CDRs of the α -chains it is observed *pan-repertoire* to pair with $\langle d_{L,\alpha CDRs} \rangle(\beta_i)$. **Bottom** Box plots of the same data with outliers outside the IQR omitted. Light yellow lines and crosses are medians and means, respectively. Grey numbers above box plots indicate the number of observations in each column. **B** The same as **A**, but where each β -chain β_i is plotted by the the number of *pan-repertoire* repertoire occurrences in the OTS $N_r(\beta_i)$ instead of by its instances. **C,D** are the same as **A,B**, but limited to the α CDR3 Levenshtein distance $\langle d_{L,\alpha CDR3} \rangle(\beta_i)$.

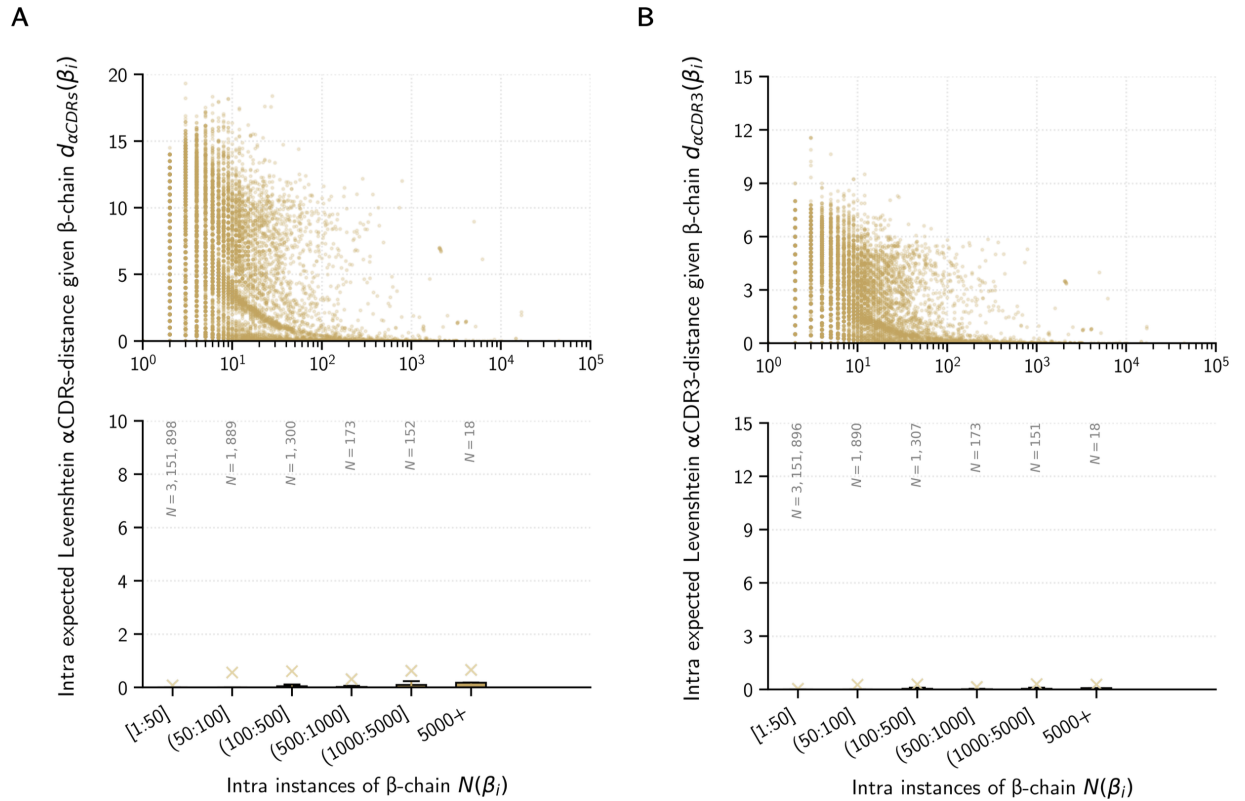


Figure 7: Intra-repertoire expected Levenshtein distance between α -chains given the β -chain. **A Top** Scatter plot, where each β -chain β_i is plotted by β_i is plotted by the its number of *pan-repertoire* instances in the OTS $N(\beta_i)$ and by the expected Levenshtein distance between the CDRs of the α -chains it is observed *pan-repertoire* to pair with $\langle d_{L,\alpha CDRs} \rangle(\beta_i)$. **Bottom** Box plots of the same data with outliers outside the IQR omitted. Light yellow lines and crosses are medians and means, respectively. Grey numbers above box plots indicate the number of observations in each column. **B** The same as **A**, but limited to the α CDR3 Levenshtein distance $\langle d_{L,\alpha CDR3} \rangle(\beta_i)$. This figure can be interpreted as an *intra-repertoire* version of [Figure 6A,C].

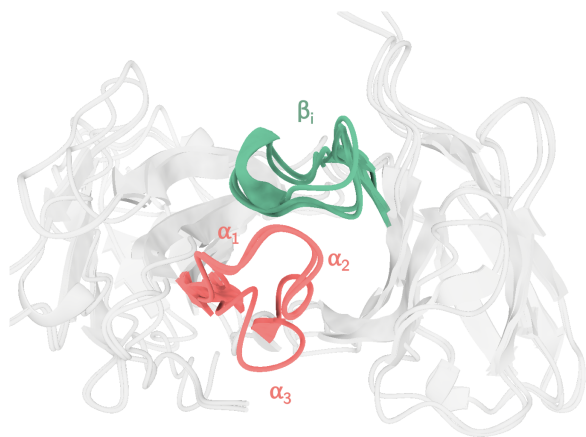


Figure 8: Structural differences in the CDR3 between distinct α -chains observed to pair with the same β -chain β_i . The CDR3 backbone of the β -chain is coloured green, and the CDR3 backbones of the three respective α -chains are coloured in red.

between α -chain partners. It was found that the α -partners of the same β -chains sampled from two repertoires on average differed in about 10% of CDR residues, and could easily differ by more than 50%, which would be expected to result in the differential recognition of distinct epitopes. As a proof-of-concept, structural divergence was demonstrated between the α -chain partners of a specific highly-sampled β -chain in predicted TCR structures.

Overall, the findings of this study suggest that β -chains in isolation are imperfect proxies for the epitope specificity that characterises the information content of TCRs. The same β -chain may carry different immunological signals in different individuals or even within the same individual, making extrapolation less reliable. Relying solely on β -chain sequences may conflate TCRs with distinct epitope specificities, potentially introducing noise, ambiguity or error into immunological repertoire-level models that impose a systematic ceiling on model performance.

The information lossiness of β -chains does not invalidate β -only sequencing or modelling efforts, but it motivates the development of correction frameworks or uncertainty quantification in diagnostic models that acknowledge the demonstrated α -chain pairing ambiguity. This study encourages a more paired-chain centric mindset that increasingly recognises α - β -chains as the ground truth and β -only approaches as approximations with quantifiable uncertainty, rather than as the default. As paired-chain sequencing throughput scales and costs fall, the marginal value of capturing both chains ought to be weighed against the low-throughput bottleneck.

This study is significantly limited by the undersam-

pling of TCR repertoires imposed by paired-chain sequencing, which simultaneously biases towards lower diversity estimates and makes these less robust. Additionally, the repertoires in the OTS are imperfectly annotated and derive from heterogeneous backgrounds; they are a mix of sample sizes, studies, disease labels, T cell types, and subjects, which may conflate genuine pairing multiplicity with annotation noise, inflating diversity estimates. However, with more repertoire and deeper paired-chain sequencing data, many of these biases are expected to be resolved.

Future work should focus on establishing a stronger proxy for functional divergence than Levenshtein distance, which is agnostic to the physicochemical properties of amino acids, positional importance, the structural complexity of binding, and the significant effect of chain lengthening or shortening. Ideally, a functional divergence measure should be based on the binding interface of predicted structures. Epitope-label databases, such as VDJdb [31] or IEDB [32], could help to establish a baseline for thresholding when a functional divergence measure can distinguish whether two different α -chain pairings bind distinct epitopes.

Additionally, it would be interesting to train single-chain β -models on β -repertoires with varying Simpson indices, to capture the effect of known pairing promiscuity on model specificity.

Further investigation into the β -chain identified as possibly 'dominant', 'promiscuous', or 'universal' (see Results 3.5; paragraph on [Figure S6]) could perhaps lead to the discovery of molecular TCR strategies that allow β -chains to be classified into distinct pairing diversity regimes.

Finally, building sophisticated Monte Carlo-based null distributions for different diversity hypotheses in the OTS would enable a more formalised and robust framework for quantifying true observed α - β -pairing diversity.

5 Data availability

The code used to perform the analyses and figures in this study is available at the public Github repository [frederikespersen/tcr-ab-pairing-diversity](https://github.com/frederikespersen/tcr-ab-pairing-diversity). The OTS data needed to run the analysis in the `main.ipynb` notebook can be downloaded by running the shell script `src/scripts/bulk_download.sh`.

6 Acknowledgements

I would like to thank the Oxford Protein Informatics Group (OPIG), Charlotte Deane, and all the OPIGlets for welcoming me in their group.

I extend a special thanks to my supervisors during my stay, Marius Urbonas and Nele P. Quast, for their supervision, cooperation and valuable insights, without which this project would have never been realised.

Finally, I extend a thanks to my UCPH supervisor, Amelie Stein, for letting me take on the project.

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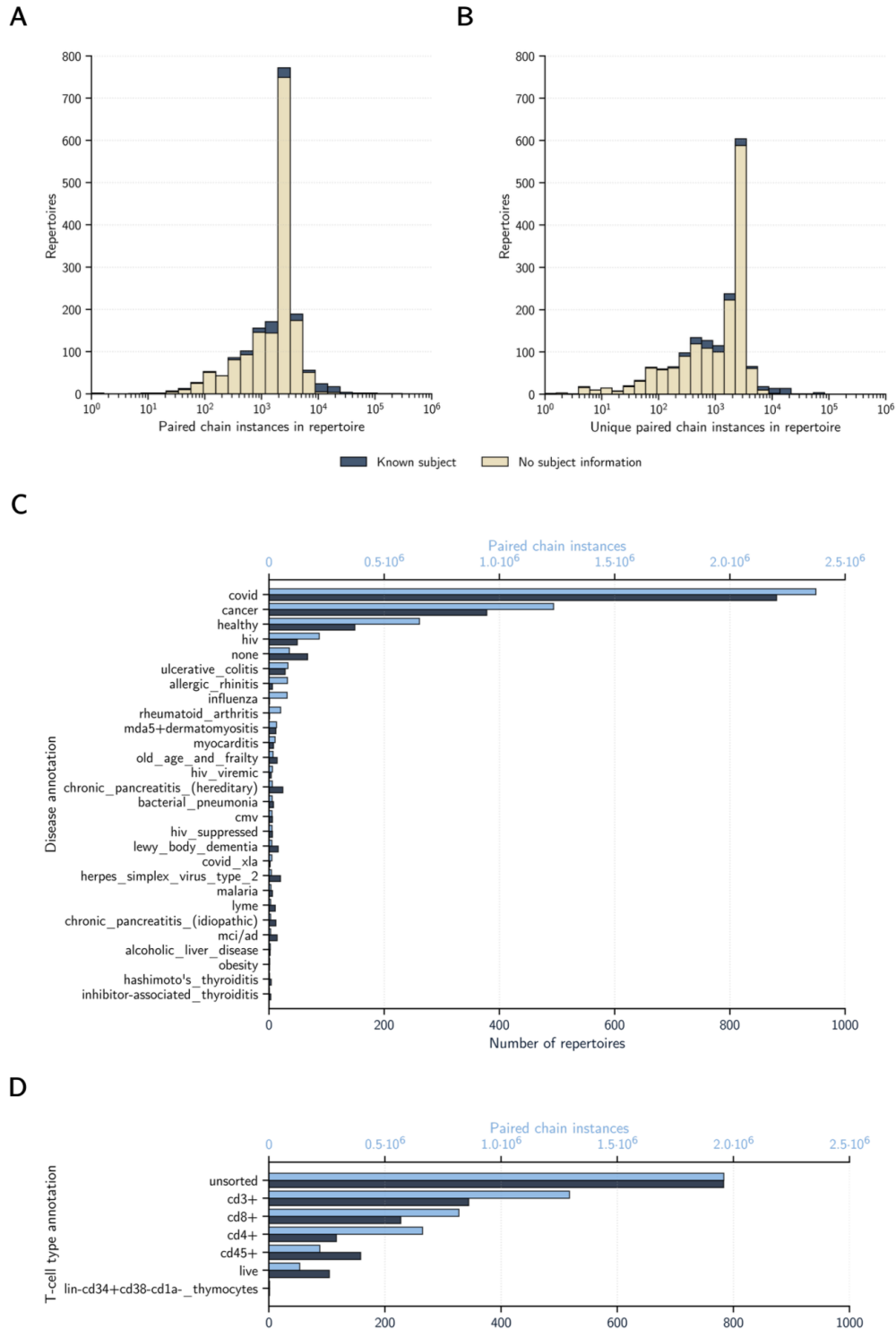
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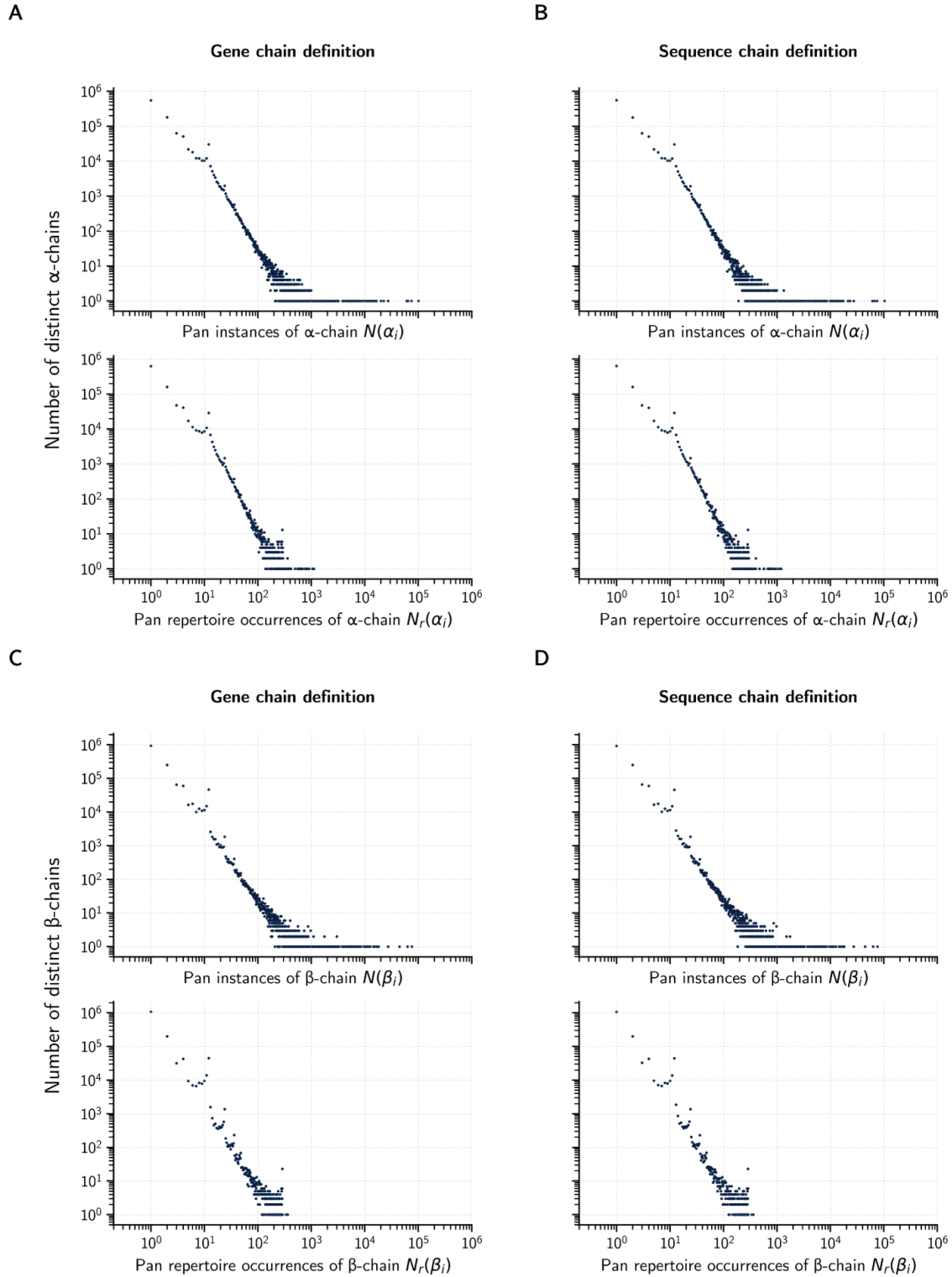
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Supplementary sequences for [Figure 8]

- β_i : DAGVTQSPTHLIKTRGQQVTLRCSPKSGHDTVSWYQQALGQGPQFIFQYEEEEERQRGNF
PDRFSGHQFPNYSSSELNVNALLGDSALYLCASSFGQSSTYGYTFGSGTRLTVV
- α_1 : LAKTTQPISMDSYEGQEVNITCSHNNIATNDYITWYQQFPSQGPRFIIQGYKTKVTNEVA
SLFIPADRKSSSTLSLPRVSLSDTAVYYCLVGDMQAGTALIFGKGTTLSVSS
- α_2 : LAKTTQPISMDSYEGQEVNITCSHNNIATNDYITWYQQFPSQGPRFIIQGYKTKVTNEVA
SLFIPADRKSSSTLSLPRVSLSDTAVYYCLVGDRDQAGTALIFGKGTTLSVSS
- α_3 : KDQVFQPSTVASSEGAVVEIFCNHSVSNAYNFFWYLHFGCAPRLLVKGSKPSQQGRYNM
TYERFSSSLILQVREADAAVYYCAVEDGRDDKIIFGKGTRLHILP

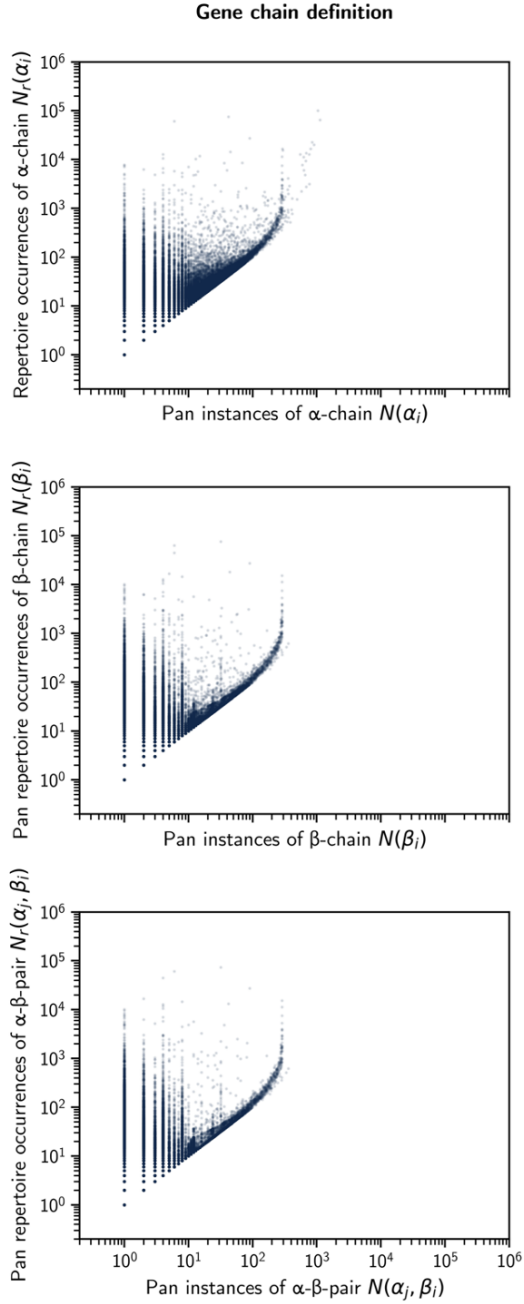


Supplementary Figure 1: Characterisation of repertoires curated from the OTS. **A,B** The distribution of repertoire sizes measured by either the total number of or the number of unique sequences, respectively, in each repertoire. Stacked bars are coloured by whether or not the repertoire is an aggregation of data units that share Subject annotation. **C,D** The distribution of disease or T-cell type annotation labels, respectively, per repertoire. The upper light blue bars indicate the number of sequence instances and the lower dark blue bars indicate the number of repertoires with each label.

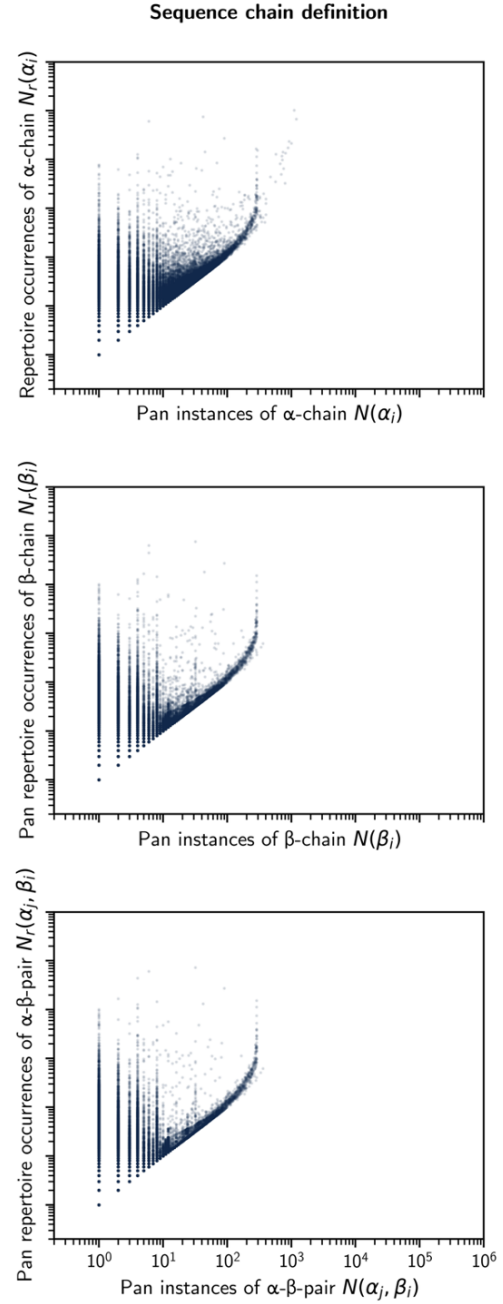


Supplementary Figure 2: Chain observations in the OTS appear to be distributed according to a power law. **A Top** Scatter plot, where each point denotes the number of α -chains that are observed a certain number of instances, using the *gene* chain definition. **Bottom** As above, but with repertoire occurrences instead of instances. **B** The same as A, but using the *sequence* chain definition. **C,D** are the same as **A,B**, but for β -chains.

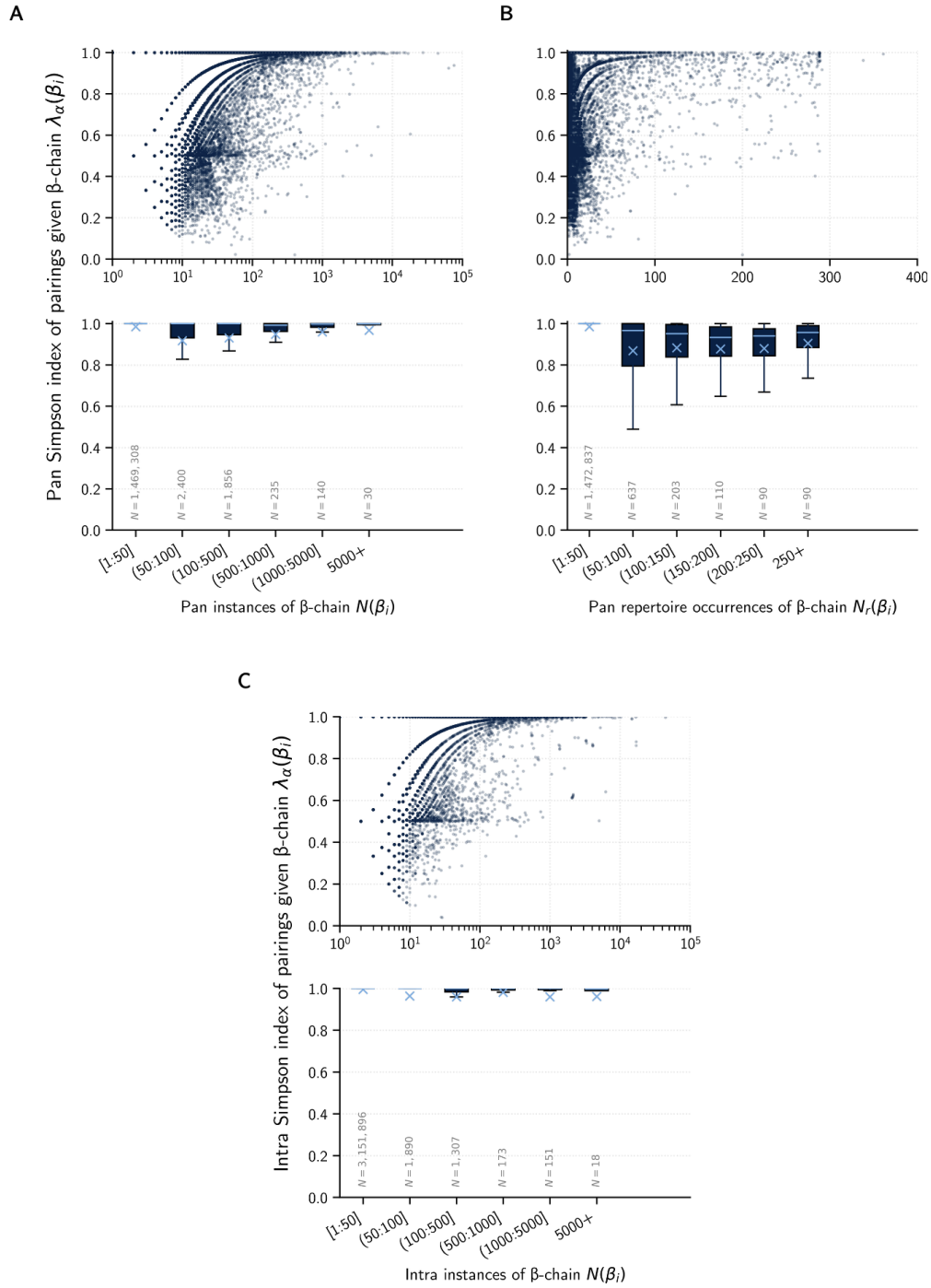
A



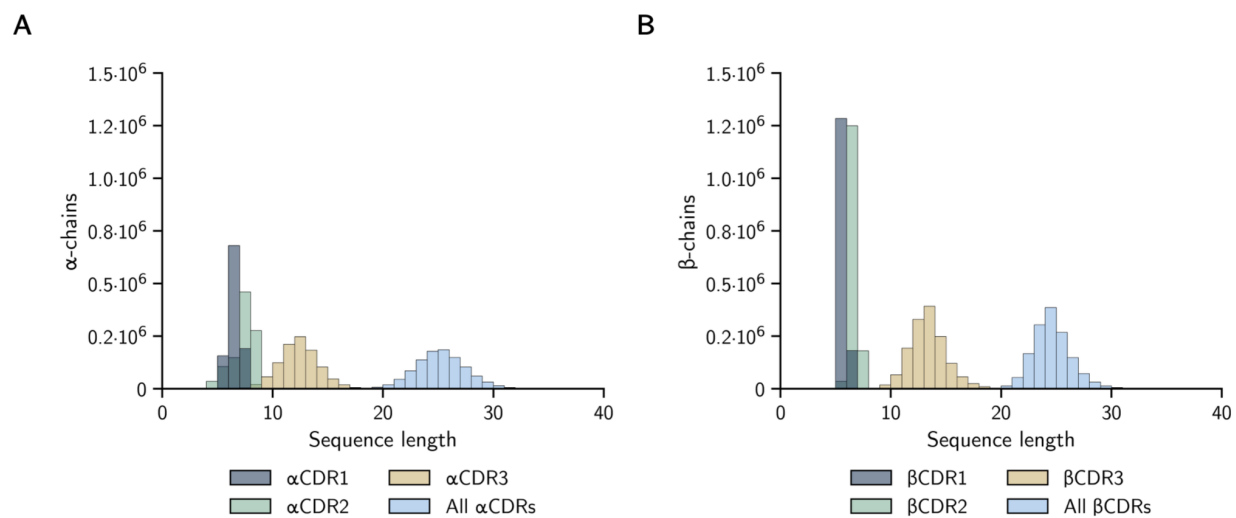
B



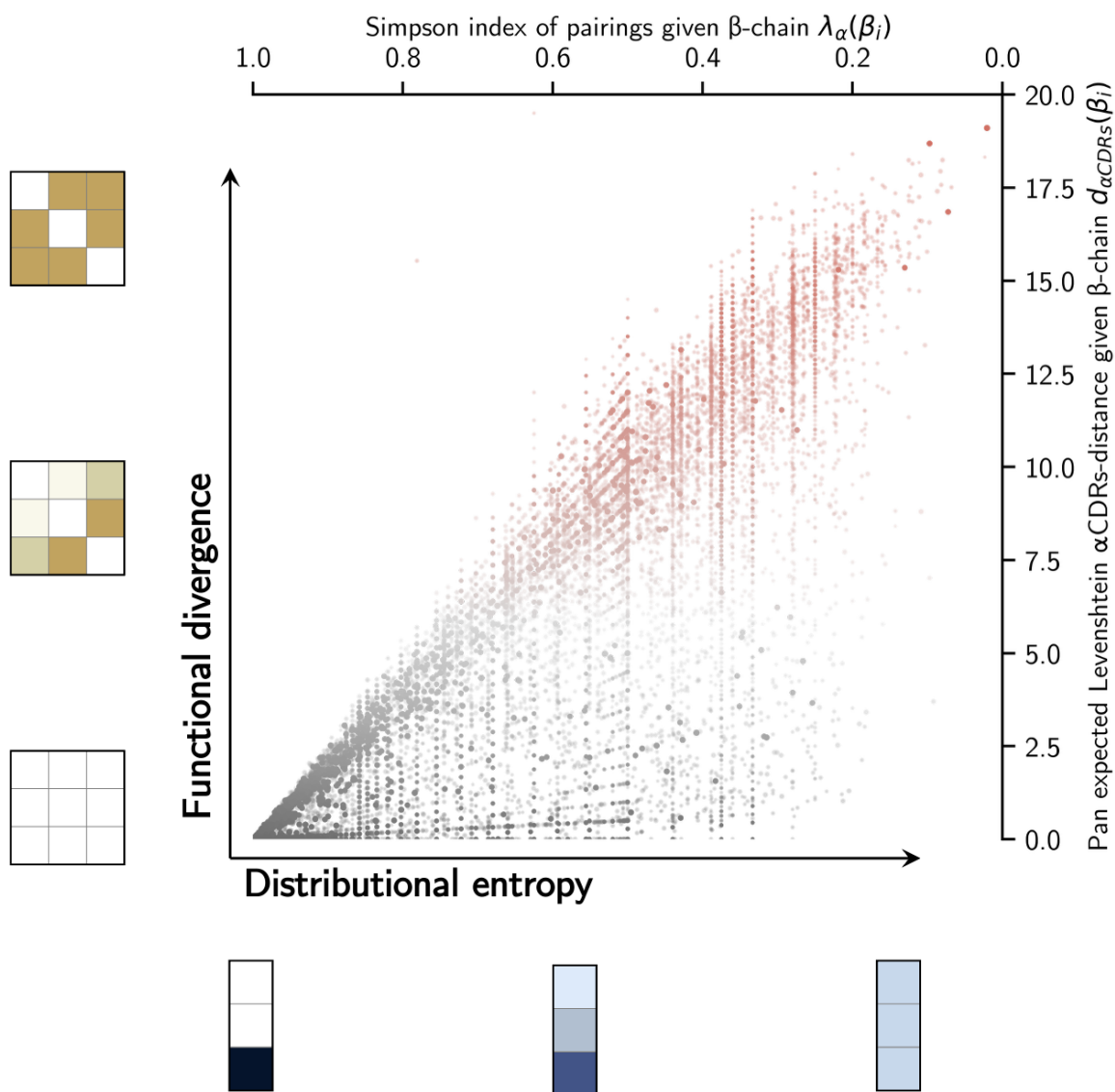
Supplementary Figure 3: Relationship between repertoire occurrences and chain instances. **A Top** Scatter plot, where each α -chain α_i is plotted by the its number of pan-repertoire instances $N(\alpha_i)$ and its number of pan-repertoire repertoire occurrences $N_r(\alpha_i)$, using the *gene* chain definition. **Middle** As above, but for β -chains. **Bottom** As above, but for α - β -pairs. **B** The same as **A**, but using the *sequence* chain definition.



Supplementary Figure 4: The Simpson index of α -chain pairing $\lambda_\alpha(\beta_i)$. **A Top** Scatter plot, where each β -chain β_i is plotted by the its number of *pan-repertoire* instances in the OTS $N(\beta_i)$ and by its *pan-repertoire* Simpson index of α -chain pairing $\lambda_\alpha(\beta_i)$. Triangles indicate out-of-axis-limits points. **Bottom** Box plots of the same data with outliers outside the IQR omitted. Light blue lines and crosses are medians and means, respectively. Grey numbers above box plots indicate the number of observations in each column. **B** The same as **A**, but where each β -chain β_i is plotted by the the number of *pan-repertoire* repertoire occurrences in the OTS $N_r(\beta_i)$ instead of by its instances. **C** The same as **A**, but *intra-repertoire*.



Supplementary Figure 5: CDR sequence lengths. **A** The distribution of CDR sequence lengths amongst α -chains for CDR1, CDR2, CDR3, and the total sequence length of all CDRs. **B** The same as **A**, but for β -chains.



Supplementary Figure 6: Observed distributional entropy and functional divergence. Scatter plot, where each point is a β -chain plotted by its *pan-repertoire* Simpson index of pairing $\lambda_\alpha(\beta_i)$ and its *pan-repertoire* expected combined α CDR Levenshtein distance $\langle d_{L,\alpha CDRs} \rangle(\beta_i)$. The size of points is scaled by the repertoire occurrences $N_r(\beta_i)$ of each β -chain. Points are coloured arbitrarily by their distance to the top right corner to highlight domains of pairing diversity. The plot is placed next to the axes from the conceptual [Figure 2B] to demonstrate a data-based realisation of the concept of simultaneous distributional entropy and functional divergence between α -chain pairings for specific β -chains. While most density is concentrated around (0,1), the plot nonetheless demonstrates that biologically diverse α - β -chain pairing does occur.